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(1) Research Program/Project/Study Title: STATUS OF WILD ORCHIDS IN THE MOUNTAIN RANGES OF PILAR, CAPIZ: BASIS FOR CONSERVATION	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:		
Component Leader:		
Academic Rank:		

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	PILAR
		Province:	CAPIZ
		Region:	6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Environmental Science	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Abstract

Orchids are highly appreciated because of their sheer beauty. It is the most evolve of all flowering plants, they are very specific and need optimum conditions to thrive in a given ecosystem. If orchids are present in anecosystem, this is an indicator of a healthy, functioning ecosystem. The problem is not just the loss of species, but also a loss of the genetic diversity. Thus this study was conducted with the aim of assessing the abundance, endemism, and conservation status of wild orchids in the mountain ranges of Pilar, Capiz. This study was conducted at 3 mountainous area in Piar, namely Mt. Agudo, Mt. San Silvestre and Mt. Tabun-aca. The orchids was surveyed using walk transect method and opportunistic sampling technique following a trail. There were 7 species of wild orchids belonging to 6 genera recorded in the mountain ranges of Pilar. *Spathoglottis plicata* Blume is the most abundant species by *Hetaeria oblongifolia* Blume, *Dendrobium crumentarum* Sw, *Vandopsis lissochiloides*, *Dendrobium anosmum* Lindl, *Grammatophyllum scriptum* and *Phalaenopsis amabilis*. Three species were found to be widespread endemic, 3 were found to be Philippine endemic and 3 were endangered species, 3 vulnerable and 1 nuetralized

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The orchid family (Orchidaceae) is the second largest family of flowering plants, with about 880 genera and some 26,000 species distributed nearly worldwide. Many are grown as ornamentals for their showy flowers, and several are of economic importance as the source of the flavoring vanilla and the most threatened group of plants globally and in the Philippines. In the Philippines, 97 species are already threatened which represents ~ 8.1% of the orchid species recorded in the country (DENR 2017).

Habitat destruction and unsustainable harvesting are the major global threats to orchids. Habitat fragmentation negatively impacts the species richness of orchids. and their highly specialized habit makes orchids more vulnerable to global environmental changes. Orchids are also highly traded mainly for horticulture, medicine, and food. Despite being regulated under the Convention on International Trade in Endangered Species, many of the species are still being collected and traded undocumented negatively affecting wild populations worldwide.

The increasing pressure on wild orchid populations put orchids at a group of urgent conservation concerns. Despite the new orchid species discoveries in recent years, diversity and distribution studies are still severely lacking. This gap may pose conservation challenges for this highly threatened plant family, as assemblage-level studies may serve as baselines for future monitoring programs in Philippine montane ecosystems, particularly for the highly collected and traded orchids. Distribution studies also help inform species conservation assessments (e.g. IUCN Red List of Threatened Species and Philippine Red List), as these provide information on the species' extent of occurrence and geographic range.

This study was conducted to assess the status of wild orchids in the mountain ranges of Pilar, Capiz for crafting a possible policy for conservation. Specifically, it aimed to 1) Identify the different species of wild orchids found in the mountain ranges of Pilar, Capiz; 2) determine the abundance of different species found in the mountain ranges of Pilar, Capiz; 3) determine the endemism of the different species found in the mountain ranges of Pilar, Capiz and; 4) determine the conservation status of the different species of wild orchids found in the mountain ranges of Pilar, Capiz.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Research Methodology

Study Area

Three mountainous barangay in the municipality of Pilar were selected as the study area. Mt. Agudo is located in Brgy. Yating Pilar, Capiz with an elevation of 1014ft above sea level. During World War II, the Americans built an antenna on top of one of these mountains that is now called Radar 1, 2, and 3. Hinulugan mountains is located in Brgy. Hinulugan, Pilar, Capiz. This is where Hinulugan Falls can be found. Is located in Brgy. The mountain of San Silvestre is located in Brgy. San Silvestre, Pilar, Capiz. This is where kawa-kawa falls is located,.

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Data Gathering

This study surveyed the abundance of wild orchids in the mountains of Brgy. San Silvestre, Tabun –acan, and Yating using a transect walk technique and opportunistic sampling following a trail from base up to the peak of the mountain from September to November 2023. The species found visually along the trail were photographed for conservation purposes and extensive examinations of understory and midstory layers of the forest for ground and epiphytic orchids were done with the caution that orchid species beyond 4 m above the ground might not have been recorded due to inaccessibility or observer bias. Both land and Epiphytic orchids, with or without flowers were enumerated and recorded.

Enumeration and Identification of Specimen

All species were enumerated and identified based on photographic guides, expert consultations through the online plant identification platform of the Co's Digital Flora of the Philippines and the Phil.plant.gov. ph. No collection of voucher specimens for conservation because many of these species are already being collected for ornamental use.

Assessment for endemism and conservation status

Conservation status of the species collected were based on the IUCN Red List of Threatened of Plants 2013 and CO.s Digital Flora of the Philippines. Endemism whether its Capiz or Panay Island Endemic (CE), Philippine Endemic (PE) or Widespread Species (WS) was based from published various floras, monographs, online databases and herbarium records.

Abundance

The abundance was measured using Simpson's Diversity Indices a formula introduced by Curtis and McIntosh (1950).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Species of Wild Orchids found in the Mountain Ranges of Pilar

There were 7 species of wild orchids belonging to 6 genera were found mountain ranges of Pilar, Capiz. These were : *Grammatophyllum scriptum*, *Dendrobium crumentarum* Sw, *Dendrobium anosmum* Lindl, *Spathoglottis plicata* Blume, *Hetaeria oblongifolia* Blume, *Vandopsis lissochiloides* and *Phalaenopsis amabilis*. This implied that only few species were found in the montain ranges of Pilar. This is because of the excessive collection of orchids for ornamental use or for sale.

Betanio, J. M. G, et. Al (2018) stated that orchids are one of the most diverse yet threatened plant families in the world and the Philippines. Despite the new species discoveries of orchids in the country in recent years, diversity and distribution studies are still severely lacking. Here, we provide a list of orchid species found among different forest types in Mount Busa and provide information on the species' altitudinal and vertical distributions and ornamental use. We recorded 108 species of orchids from 51 genera, of which 53 species are endemic to the Philippines and 15 species are known only to occur on Mindanao island. Species composition differed among forest types, with the MASLEF (mixed advanced secondary lowland evergreen forest) hosting the highest number of orchid species recorded, followed by the UMF (upper montane forest), LMF (lower montane forest), and MESLEF (mixed early secondary lowland evergreen forest). Our results suggest a relatively rich and distinct orchid diversity among different forest types in Mount Busa that reinforces the high conservation value of the mountain range.

Table 1. Species of orchids found in three study area

Name of Species	Brgy. San Silvestre	Brgy. Tabun-Acan	Brgy. Yating
1. <i>Grammatophyllum scriptum</i>	1	5	7
2. <i>Dendrobium crumentarum</i> Sw	3	10	12

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3. Dendrobium anosmum Lindl	0	5	9
4. Spathoglottis plicata Blume	2	15	28
5. Hetaeria oblongifolia Blume	2	9	18
6. Vandopsis lissochiloides	0	5	14
7. Phalaenopsis amabilis	0	0	3



Grammatophyllum scriptum



Dendrobium crumentarum Sw



Dendrobium anosmum Lindl



Spathoglottis plicata Blume



Hetaeria oblongifolia Blume



Vandopsis lissochiloides



Phalaenopsis amabilis

Species Abundance of Wild Orchids found in the Mountain Ranges of Pilar

Of the 7 species found in the mountain ranges of Pilar, *Spathoglottis plicata* Blume is the most abundant species with abundant rate of 15 followed by *Hetaeria oblongifolia* Blume with 9.67 abundant rate, *Dendrobium crumentarum* Sw with 8.33, *Vandopsis lissochiloides* with 6.33,

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Dendrobium anosmum Lindl with 4.67, Grammatophyllum scriptum with 4.66 and Phalaenopsis amabilis with an abundant rate of 1. This implied that Brgy. Spathoglottis plicata Blume is the most abundant species that is also present in 3 sampling areas.

Research in forest canopies has been restricted by access difficulties for a long time, however in the last 20 years techniques to provide access into tree canopies have been developed, now opening a field of biology to more scientists. This study uses rope techniques for accessing the canopy in a primary rainforest system in Borneo, Malaysia, to study the occurrence and patterns of orchid distribution in Parashorea malaanonan dipterocarp trees. Five trees were sampled, a total of 42 orchids from 8 genera were present. Data on branch height, distance from bole, position of the orchid on branch, substrate and associations were recorded and then analysed to find any noteworthy interactions. Significant relationships were found between many of the variables. Branch height was found to correlate with substrate, association, genera, species and position on branch, implying that it is a major contributor to orchid variation. Substrate showed a relationship with both genera and species, and non-uniform distribution across the five trees sampled. Association and position on branch were found to have an inverse relationship, as were genera and tree. However, this study could not establish definitive reasons for the relationships between variables and patterns in orchid diversity and abundance(O'Malley, K. 2009).

Table 2. Species abundance of wild orchids in the mountain ranges of Pilar

Name of Species	No. of Species/Brgy.			Total	Abundance Rate
	Brgy. San Silvestre	Brgy. Tabun-acan	Brgy. Yating		
Grammatophyllum scriptum	1	5	7	13	4.33
Dendrobium crumentarum Sw	3	10	12	25	8.33
Dendrobium anosmum Lindl	0	5	9	14	4.67
Spathoglottis plicata Blume	2	15	28	45	15
Hetaeria oblongifolia Blume	2	9	18	29	9.67
Vandopsis lissochiloides	0	5	14	19	6.33
Phalaenopsis amabilis	0	0	3	3	1
Total	8	49	91		

Species Endemism of Wild Orchids found in the Mountain Ranges of Pilar

Four species were found to be Wide Spread or can be located in different country or continents. These were Grammatophyllum scriptum, Spathoglottis plicata Blume, Hetaeria oblongifolia Blume and Vandopsis lissochiloides and only three were found to be endemic in the Philippines. These were Dendrobium crumentarum Sw, Dendrobium anosmum Lindl and Phalaenopsis amabilis. This implied that there were only few species that is endemic in the Philippines were found in the mountain ranges of Pilar.

According to Pedersen, H. A.A et. Al (2019) phylogenetic hypothesis for the genus Dendrochilum s.l. (including Bracisepalum) is based on nucleotide sequences of the nuclear ribosomal ITS and two plastid regions (matK, ycf1). The trees based on parsimony analysis of the nrITS and plastid partitions, respectively, are largely congruent, the only strongly supported conflict being the exact position of D. arachnites. Maximum likelihood analysis of the combined molecular matrix resulted in a tree with very high topological congruence to the parsimony consensus tree and to a fossil-calibrated Bayesian inference tree that was used for dating. Our results confirm that monophyly of Dendrochilum is only supported if Bracisepalum is included in the former. They also suggest that only two subgenera (Dendrochilum, Platyclinis) should be recognized, whereas all other subgenera and sections previously proposed on a morphological basis should be treated as synonyms of subg. Platyclinis. Indeed, mapping of five vegetative and nine floral characters of alleged systematic value, using DELTRAN and ACCTRAN optimization, demonstrates pronounced morphological homoplasy. Molecular time calibration dates the origin of the genus to the Miocene and contradicts the hypothesis that most extant Dendrochilum species did not evolve until the

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Holocene. Inferred phylogenetic relationships between extant species, in combination with previously identified hotspots of narrow endemism, suggest that species endemism in *Dendrochilum* is predominantly centric.

Table 3. Species endemism of wild orchids

Name of Species	Endemism
<i>Grammatophyllum scriptum</i> (Blume, <i>Rumphia</i> 4 (1848) 48	(WS) Wide Spread
<i>Dendrobium crumentarum</i> Sw (Cootes, Orch. Philip. (2001) 73)	(PE) Philippine Endemic
<i>Dendrobium anosmum</i> Lindl (Cootes, Orch. Philip. (2001) 66)	(PE) Philippine Endemic
<i>Spathoglottis plicata</i> Blume (Lubag-Arquiza, Orchid Digest 70(3) (2006)	(WS) Wide Spread
<i>Hetaeria oblongifolia</i> Blume (Fl. Jav. Orch. (1858) 85)	(WS) Wide Spread
<i>Vandopsis lissochiloides</i> (Ames in Merr., EPFP 1 (1925) 427)	(WS) Wide Spread
<i>Phalaenopsis amabilis</i> (Ames in Merr., EPFP 1 (1925) 410)	(PE) Philippine Endemic

Conservation Status of Wild Orchids found in the Mountain Ranges of Pilar

There were 3 endangered species recorded in the mountain ranges of Pilar. These were *Dendrobium crumentarum* Sw, *Dendrobium anosmum* Lindl and *Phalaenopsis amabilis*. Three species were vulnerable: *Grammatophyllum scriptum*, *Spathoglottis plicata* Blume and *Hetaeria oblongifolia* Blume and *Vandopsis lissochiloides* is neutralized.

As stated by Panal CLT, Opiso JG, Opiso G. (2015) conservation status of the Family Orchidaceae in Mt. Sinaka, Arakan, North Cotabato, Philippines. *Biodiversitas* 16: 213-224. This study determines the conservation status of the family Orchidaceae, as present on Mt. Sinaka, Arakan, North Cotabato, Philippines. A thorough survey and alpha taxonomy was done, from base to peak of the mountain. Identification of the specimens and assessment of their conservation status was based on the IUCN Red List of Threatened Plants 2013.2 and National List of Threatened Philippine Plants of Fernando et al. (2008). Based on the result conducted on October 2013-March 2014, out of 59 identified species found in the area, 12 species are widespread, 22 are endemic, 1 vulnerable, 1 critically endangered (*Paphiopedilum adductum*), 1 endangered (*Corybas* sp.), 2 least concerned species, and 20 unassessed species (not yet assessed by the IUCN). It has been also noted that there are probably some new species, which need thorough study for further identification. The result calls for a desperate need for conservation. Facts from this study helps in addition to the existing wildlife conservation of flora in Mt. Sinaka and to the other forested mountains in Mindanao, the Philippines.

Table 4. Conservation Status of wild orchids

Name of Species	Conservation Status
<i>Grammatophyllum scriptum</i>	Vulnerable
<i>Dendrobium crumentarum</i> Sw	Endangered
<i>Dendrobium anosmum</i> Lindl	Endangered
<i>Spathoglottis plicata</i> Blume	Vulnerable
<i>Hetaeria oblongifolia</i> Blume	Vulnerable
<i>Vandopsis lissochiloides</i>	Nuetralize
<i>Phalaenopsis amabilis</i>	Endangered

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

There were 7 species of wild orchids belonging to 6 genera recorded in the mountain ranges of Pilar. *Spathoglottis plicata* Blume is the most abundant species by *Hetaeria oblongifolia* Blume, *Dendrobium crumentarum* Sw, *Vandopsis lissochiloides*, *Dendrobium anosmum* Lindl, *Grammatophyllum scriptum* and

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Phalaenopsis amabilis. Three species were found to be widespread endemic, 3 were found to be Philippine endemic and 3 were endangered species, 3 vulnerable and 1 neutralized.

Recommendation

1. Strict implementation of the policy of wildlife conservation.
2. Policy formulation for conservation of the existing species.
3. Application for the area to be protected by the DENR.
4. CapSU Pilar should put up center for germplasm collection for wild plants for preservation

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.) Journal

Journal and Table Coffee Book, brochure, pamphlets as supplementary instructional materials for dissemination to concerned offices. This will also be subject for copyrighted

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Betanio, J. M. G., & Buenavista, D. P. (2018). Elevational Pattern of Orchid Rarity and Endemism in Mt. Kalatungan, Mindanao Island, Philippines. *Journal of Tropical Life Science*, 8(2), 238106.

O'Malley, K. (2009). Patterns of abundance and diversity in epiphytic orchids on *Parashorea malaanonan* trees in Danum Valley, Sabah.

Pedersen, H. A., Petersen, G., Gravendeel, B., Barkman, T. J., de Boer, H., Sulistyo, B. P., & Seberg, O. (2019). Phylogenetics of *Dendrochilum* (Orchidaceae): Evidence of pronounced morphological homoplasy and predominantly centric endemism. *Taxon*, 68(6), 1173-1188.

T PANAL, C. L., OPISO, J. G., & OPISO, G. (2015). Conservation status of the Family Orchidaceae in Mt. Sinaka, Arakan, North Cotabato, Philippines. *Biodiversitas Journal of Biological Diversity*, 16(2).